

Final Analysis & Reporting

Q. Conclusion and Business Recommendation:

➤ Strengths and Weaknesses of Each Classifier

Classifier	Strengths	Weaknesses
KNN	Simple to understand, no training time, works well with small data	Slow at prediction, sensitive to scaling and noise, poor on imbalanced data
SVM (Linear)	Fast, good for linearly separable data, finds maximum margin	May underperform if data is not linearly separable
SVM (RBF)	Handles non-linear boundaries, very powerful, high accuracy	Slow on large datasets, harder to interpret, needs proper tuning
Naive Bayes	Very fast, probabilistic, interpretable, handles small data	Independence assumption is often violated in real data, may be less accurate

➤ Impact of Probability Assumptions in Naive Bayes:

- Naive Bayes assumes that all features are independent of each other. In our dataset, features like spam_keyword_score and num_urls are likely correlated (spam messages tend to have both). This violated assumption may slightly hurt Naive Bayes accuracy. However, despite this limitation, Naive Bayes often still performs surprisingly well in practice.

➤ Trade-offs Between Interpretability and Performance:

- KNN and Naive Bayes are the most interpretable - you can explain each prediction easily.
- SVM with RBF kernel may give the best accuracy but is harder to explain to a business user.
- In a security system, we may accept a complex model if it delivers better recall (catching more spam).

Conclusion:

- In this project, we built a Message Intelligence System to classify messages as Spam or Legitimate using three supervised learning algorithms: K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Naive Bayes.
- After loading and analyzing the dataset of 5200 messages, we performed necessary preprocessing including median imputation for missing values and StandardScaler for feature scaling. We split the data into 80% training and 20% testing sets.
- We trained KNN with multiple K values and found that the optimal K gave a good balance of accuracy and F1 score. SVM with both Linear and RBF kernels was tested, and the RBF kernel

generally performed well because it captures non-linear decision boundaries. Naive Bayes was fast and interpretable but slightly less accurate due to the feature independence assumption.

- For business deployment, SVM (RBF) is the recommended model as it tends to achieve the best overall performance. However, if interpretability is important (e.g., for compliance or auditing), Naive Bayes or KNN can be preferred alternatives.
- Key features that strongly predicted spam were `spam\_keyword\_score`, `num\_urls`, and `legit\_keyword\_score`. The manual Bayes' Theorem calculation confirmed that messages with URLs have a significantly higher probability of being spam, which aligns with real-world spam patterns.
- Final Recommendation: Deploy SVM (RBF) as the primary classifier for the Message Intelligence System, with a focus on optimizing recall to minimize false negatives (spam messages slipping through).